

TRANQUILITY IOT & BIG DATA SOLUTIONS

TIBS IOT GATEWAYS AND MULTIPORT WIRELESS CONTROLLERS

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IoT GATEWAY

MULTIPORT Wireless Controller

- Continuous Monitoring of Industrial Machine Vitals for Real Time Alerting
- IoT Enabling Legacy Systems through TIBS Intelligent Convertors
- Optimizing Energy Utilization through Real Time Data Acquisition and Analytics
- Cost Effective IoT enabling through Multiport Wireless Controllers (Wi-Fi, BLE, Zigbee, LoRa, LTE with GPS/GLONASS)
- Affordable Tailor Made Automation Solution Development for Manufacturing Domain
- Inbuild Security for all TIBS Products
- Real time Machine/Equipment Utilization Monitoring to Optimize Production Process Flows and Overall Cost reduction
- Retrofit Sensors onto Existing Machines and Monitor the Machine Vitals

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YOUR TRUSTED PARTNER FOR ELECTRONIC MANUFACTURING SERVICES

ONE STOP SOLUTION
 NO MINIMUM ORDER
 CLOUD BASED SERVICE

DESIGN

PCB LAYOUT

- Upload schematics and BOM, and instantly receive quotations based on the chosen layer count and capabilities ranging from 1-10 Layers.

FABRICATION

- Select Gerber files, layer count, material, surface finish, and quantity. Diac Labs ensures flawless fabrication, even for the most intricate designs.

SOURCING

ELECTRONICS COMPONENT

- Provide BOM and quantities. Our platform generates quotations along with precise delivery timelines. Genuine components traceable to their source country and featuring latest date codes.

ASSEMBLY

- Diac Labs combines automation and expertise to ensure accurate and efficient assembly. Our capabilities include fully automated SMD and SMT assembly as well as BGA assembly extending to box building.

ADVANTAGES

- OPTIMAL PRODUCTION COST
- PROVEN EXPERIENCE
- ESTABLISHED VENDOR RELATIONSHIPS
- IN HOUSE PROTO & PRODUCTION SUPPORT
- ACCELERATED TIME TO MARKET

CONTACT US

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nents used in the robot is provided below.

Servo motor SG90. A servo motor is a specialised type of motor that rotates with great accuracy and high precision. In servo motor, a control circuit continuously provides feedback on the current position of the motor shaft, enabling precise rotation. Servo motors are generally rated in kg/cm (kilogram per centimetre), with hobby servo motors rated at 3kg/cm, 6kg/cm, or 12kg/cm. This rating indicates the weight the servo motor can lift at a particular distance.

Arduino board. The Arduino microcontroller board is based on an ATmega microcontroller. It features digital input/output pins, analogue inputs, a USB connection, a power jack, an ICSP header, and a reset button.

Ultrasonic sensor HC-SR04. The HC-SR04 is distance measur-

ing sensor with a range of 2cm to 400cm. The sensor incorporates two ultrasonic transducers, one of which is a transmitter that sends ultrasonic sound pulses and the other a receiver that listens for reflected waves, essentially functioning as a sonar.

Software

The code is built using the Arduino IDE. Initially, the servo.h library is employed to drive the servo, and then pins for the ultrasonic sensor are defined. Although any free pins can be used, pins D11 and D12 of Arduino Uno are used here as Trig (2) and Echo(3) pins of the ultrasonic sensor.

The Trig pin sends ultrasonic waves, and the Echo pin receives the waves. The time difference between sending and receiving is converted into distance. Consequently, the robot can sense

someone approaching and, in response, either the servo motor moves to welcome or any actuator can replace the servo for the automatic opening of doors, windows, or garage gates.

In the setup function, the servo SIG pins of servo motors S1 and S2 are defined. Here D9 and D10 pins of Arduino are used, but any PWM pins can be utilised. In the loop function, the distance is checked, and if the distance is below the threshold value that can be set, the servo motor moves to welcome. Fig. 3 shows a snippet of the source code.

Mechanical construction

For the mechanical construction of the robot, you can design any body of your choice, purchase one from the market, or 3D print an open source robot hand like the inMoov. For simulation and testing demo